

Automatic Bottle Filling and Capping Machine using PLC

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Abstract— The goal of this paper is to design and create "Bottle filling and capping with PLC". This work gives a ton of advantages like low power utilization, low operational cost and some more. This undertaking is in light of Industrial Automation and is a tremendous application used in numerous businesses like milk industries, chemical, food, mineral water and numerous production industries. A model has been created to delineate the task. Filling is the task that is completed by a machine and this procedure is generally used in numerous industries. The filling of the bottle is controlled by using a controller known as PLC. For the conveyor system, a DC motor has been chosen for better execution and simplicity of activity. Different sensors have been utilized to identify the position of the bottle. In our undertaking we have utilize less quantities of framework henceforth the general expense has been decreased to a degree. Ladder logic is used for the programming of the PLC, which is the most broadly used also, acceptable language for the programming of the PLC. The PLC used right now an SELEC PLC which makes system more adaptable and simple to work.

Keywords: SELEC PLC, Proximity sensor, Conveyor, photo electric sensor, Ladder Logic, solenoid valve etc.

I. INTRODUCTION

Fundamentally Automation is the thing which lessens the human work and expands the profitability. At the point when the innovation expands the efficiency likewise increments. PLC acts a significant role in the automation field which will in general diminish the intricacy, expands wellbeing and cost efficient. Most significant utilization of automation is in soft drinks bottle filling and capping framework.

Our task having three areas includes the Sensing the bottle on the conveyor belt then dispensing the required amount of liquid into the bottle by the solenoid valve. Then it will move the bottle over the conveyor belt. Capping system is accomplished by utilizing dc gear engine and sensors. Ladder logic is used to control the entire procedure. Sensors are utilized to detect the bottles and perform capping operation. The whole system operation is controlled by SELEC PLC which operates on 24V DC and is a compact PLC which has a fixed number of inputs and outputs.

II. OBJECTIVES

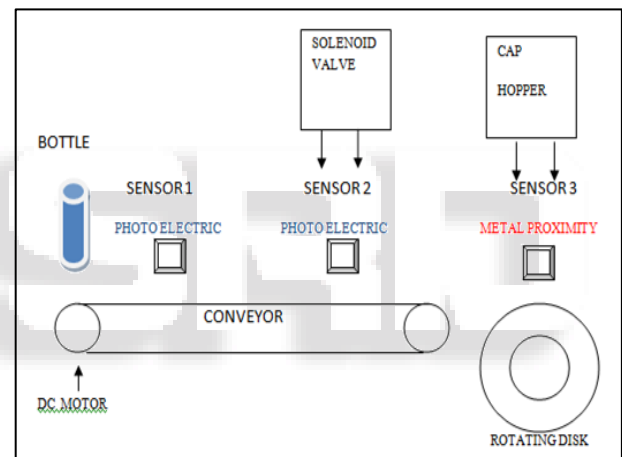
Our primary objective is design and implement the "Bottle filling and capping machine using PLC" in order to reduce the man power and increase the output as well as accuracy.

III. PROCESS ALGORITHM

- 1) STEP 1: Press the "START" Push Button.
- 2) STEP 2: Motor starts and conveyor is moving forward.

- 3) STEP 3: If sensor detects empty bottle in filling section, conveyor will stop.
- 4) STEP 4: After some delay the valve will turn "ON" and the bottle will get filled with liquid.
- 5) STEP 5: After the bottle is filled, conveyor will move forward.
- 6) STEP 6: As the bottle is moving forward, the cap will be placed by cap hopper automatically.
- 7) STEP 7: After the cap is placed the bottle will move further and the bottle will stop below the capping piston.
- 8) STEP 8: After some delay the capping piston will come down and the cap will be placed tightly.
- 9) STEP9: After the cap is placed he bottle will go further process.

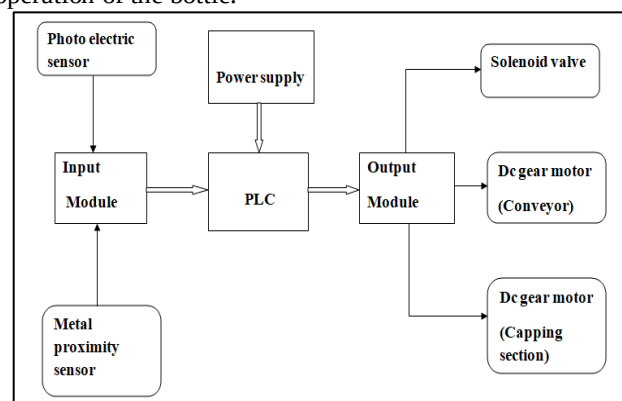
IV. PROCESS CHART



V. PROPOSED SYSTEM

Our undertaking depends on modern computerization and PLC is the core of automation. The hardware and programming are the two significant regions in our task.

The block diagram of the system is as shown in the fig. Where the block diagram consists of the following main blocks as input module, plc, solenoid valve, output module. These all blocks are help to do the filling and capping operation of the bottle.



VI. HARDWARE DESCRIPTION

Hardware part of the system consists of PLC (programmable logic controller) particularly SELEC type we have used .It is the heart of our system. Other hardware parts are photo electric sensor (2),metal proximity sensor(4), 24V SMPS , 6mm diameter solenoid valve, On-Off push buttons ,conveyor, rotating metal disc , 12V 30 RPM dc gear motor(4),24v Relay .On-Off buttons used for input devices. The PLC controls the input and output depending on program given.

A. PLC

A programmable logic controller (PLC) is an industrial digital computer which has been used for control of industrial process.



It takes the input from sensors and executes the logic programmed in its memory and generates the output to control the devices.

In this project we have used SELEC PLC for controlling the inputs and outputs. Input supply to the PLC is given through a SMPS. The rating of the SMPS is 24V DC 5Amp. The PLC we used is a compact PLC which has fixed number of inputs and outputs. In our PLC, we have 8 digital inputs and 5 digital outputs

B. Operation of PLC

The PLC is connected with some devices and it is made to run with the help of program. PLC executes the program by one instruction at a time, where, if the first input is ON then it should turn on the first output. Hence it already knows which input is on/off, from the previous process. PLC updates the status of the outputs. First the PLC checks the input status and it scans the input by user defined programming. Next the process is executed and finally it checks the output status.It continuously monitor the devices input status and output status.

C. Solenoid Valve

It is electromagnetic valve which is used to Control the flow of various types of liquids by opening and closing the valve on the Application of current .In our project we have

connected solenoid Valve to the water tank. Operating voltage is 24v.



D. Metal Proximity Sensor

An inductive proximity sensor is a non-contact electronic proximity sensor. It is used for positioning and detection of metal objects. Proximity sensor often emits an electromagnetic field or a beam of electromagnetic radiation and looks for changes in the field or return signal. In our project we are using 12V NO type sensor used.



E. Selection of PLC

While choosing a PLC, different elements ought to be considered according to the application. The significant elements that affect these criteria are:

- Number of Inputs and Outputs.
- Speed
- Communication Protocol
- Memory Size
- Software Distinctive

Different PLC'S have different types of input and output modules. Only compact type has fixed number of inputs and outputs. Expansion type modules are available to increase the I/O but it compatible only with high end modules. Some plc we have add external I/O cards to increase the input and output ports.

VII. SOFTWARE DESCRIPTION

There are varieties of languages used for programming PLC'S. But only five languages are most used in all industrial application also. These are

- Instruction list
- Flow chart
- Ladder diagram
- Functional block diagram
- Structure text

Out of these five languages ladder diagram is most used. Because it is simple and efficient.

A. Software Used and Programming

Selec Soft is used for programming all selec plc. In this software we can write a program in ladder logic format.

In our system initially on switch pressed the motor will start, when the motor start the conveyor will move in forward direction. When the photo electric sensor detect the bottle in filling section, conveyor immediately stop. Then PLC will on solenoid valve to fill water in particular time till the conveyor is stop. After the timing is expired the conveyor will move again forward, it lift the bottle from conveyor to rotating disc. When bottle perfectly placed rotating disc metal proximity sensor send the signal to controller to start motor2. Cap automatically placed at bottle head by cap hopper. When it comes to capping position rotating disc motor stop. Then two dc gear motor start and perform capping operation. After capping is finished motor will rotate and lift bottle to conveyor 2. conveyor 2 is always continuously running. It deliver the bottle to next section.

VIII. FUTURE WORK

With the help of proper and efficient components, the drawback of productivity can be overcome. Here we have used a two port valve for the filling purpose. Instead of that we can use a jet nozzle for the filling purpose since it will reduce the time of filling operation. That means more bottle will be filled and this will reduce the time and increase productivity. Also level sensor can be used for the level detection in the bottle while filling. Alarm function can be added in case of any misoperation.

IX. CONCLUSION

The purpose of this project is to develop a PLC based automatic bottle filling system plant. We gained more knowledge about various processes directly used in industries such as filling, capping etc. which are used in automation system in which we specifically learnt about Programmable Logical Controller (PLC).

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